

Example Calculate the boundary layer thickness for a jetliner.

$$V_{\infty} = 400 \text{ miles/hour } (177 \text{ m/s}) \quad (\text{Plane speed})$$

$$\nu_{\text{air}} = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$$

$$L = 5 \text{ m} \quad (\text{Length of wing in the fuselage direction})$$

$$Re_L \approx \frac{V_{\infty} L}{\nu} = \frac{(177 \text{ m/s})(5 \text{ m})}{1.5 \times 10^{-5} \text{ m}^2/\text{s}} = 5.9 \times 10^7$$

$$\delta \approx \frac{L}{\sqrt{Re_L}} = \frac{5 \text{ m}}{\sqrt{5.9 \times 10^7}}$$

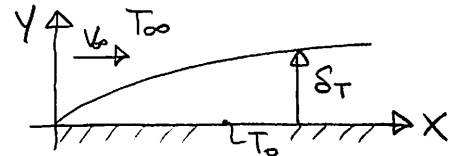
$$\boxed{\delta = 0.65 \text{ mm}} \Rightarrow \text{Less than 1 mm thick! Very hard to see.}$$

END OF LECTURE 9

Heat Transfer

Note we are looking for heat transfer on the external surface

$$h = \frac{q''_{y=0}}{\Delta T} = \frac{q''_{y=0}}{T_0 - T_{\infty}} = ?$$



For a flat plate, laminar flow: $k, \rho, \mu, C_p = \text{constant}$

Writing out our energy equation for a fluid element, which is the heat equation but extended to allow for fluid motion

$$\underbrace{u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}}_{\text{Convection of heat}} = \underbrace{\alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)}_{\text{Conduction of heat}} \Rightarrow \text{Usually we have } \nabla^2 T = 0$$

$$\text{or } \nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

0 since steady.

For a boundary layer at $T_0 = T_{\text{wall}} = \text{constant}$

$$\frac{\partial^2 T}{\partial x^2} \ll \frac{\partial^2 T}{\partial y^2}$$

Boundary conditions: $T(y=0) = T_0$
 $T(y \rightarrow \infty) = T_{\infty}$

Non-dimensionalizing: Let $\Theta = \frac{T - T_0}{T_{\infty} - T_0}$

$$u \frac{\partial \Theta}{\partial x} + v \frac{\partial \Theta}{\partial y} = \alpha \frac{\partial^2 \Theta}{\partial y^2} \Rightarrow \text{Note the similarity to the boundary layer eqns.}$$

Previously we had:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \Rightarrow \text{Hydrodynamic boundary layer}$$

$$u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \alpha \frac{\partial^2 \theta}{\partial y^2} \Rightarrow \text{Thermal boundary layer}$$

$$\begin{aligned} \text{B.C.'s : } \theta(y=0) &= 0 & \bar{u}(y=0) &= 0 \\ \theta(y \rightarrow \infty) &= 1 & \bar{u}(y \rightarrow \infty) &= 1 \\ \frac{\partial \theta}{\partial y} \Big|_{y \rightarrow \infty} &= 0 & \frac{\partial \bar{u}}{\partial y} \Big|_{y \rightarrow \infty} &= 0 \end{aligned} \quad \text{vs.}$$

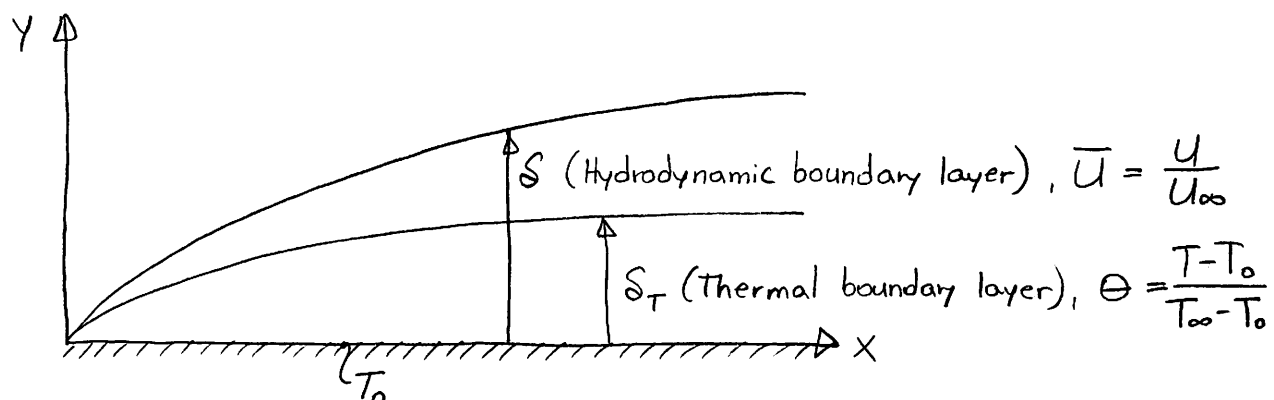
So our solution is already solved. Just use the b.l. soln's.

Here, we can define a usefull quantity called the Prandtl number

$$\boxed{Pr = \frac{\nu}{\alpha} = \frac{\text{viscous diffusion rate}}{\text{thermal diffusion rate}}}$$

If $Pr = 1$, $\theta = \bar{u}$, $\delta = \delta_T \Rightarrow$ Hydrodynamic & thermal boundary layers are same.

But usually $\nu \neq \alpha$. If $\nu > \alpha$, the hydrodynamic boundary layer is thicker since you transfer momentum more efficiently than thermal energy.



Now lets solve the thermal boundary layer equation.

If $Pr \neq 1$; $\theta(\eta)$, $\eta = y \sqrt{\frac{V_\infty}{xU}}$

$$\frac{\partial \theta}{\partial x} = \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \quad \frac{\partial \eta}{\partial x} = \frac{\partial \eta}{\partial y} \sqrt{\frac{V_\infty}{xU}} = \frac{\partial \eta}{\partial y} \frac{1}{\sqrt{x}}$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y}$$

$$\frac{\partial^2 \theta}{\partial y^2} = \frac{\partial^2 \theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2$$

Now substituting back into our energy equation PDE,

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F Pr \frac{\partial \theta}{\partial \eta} = 0, \text{ note } F = \int_0^\eta \phi d\eta = F(\eta)$$

$$\bar{u} = \phi(\eta) = \frac{y}{V_\infty} = F'$$

To solve we usually integrate but here we can use a trick.
Let:

$$Pr^{2/3}, F(\eta) = F(\eta^*) ; \quad \eta^* = \eta Pr^{1/3}$$

$$\frac{\partial \eta^*}{\partial \eta} = Pr^{1/3}$$

Now our PDE becomes:

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{1/3} \frac{\partial \theta}{\partial \eta} = 0$$

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{2/3} \underbrace{\frac{\partial \theta}{\partial \eta Pr^{1/3}}}_{\partial \eta^*} = 0$$

Multiplying through by $Pr^{-2/3}$

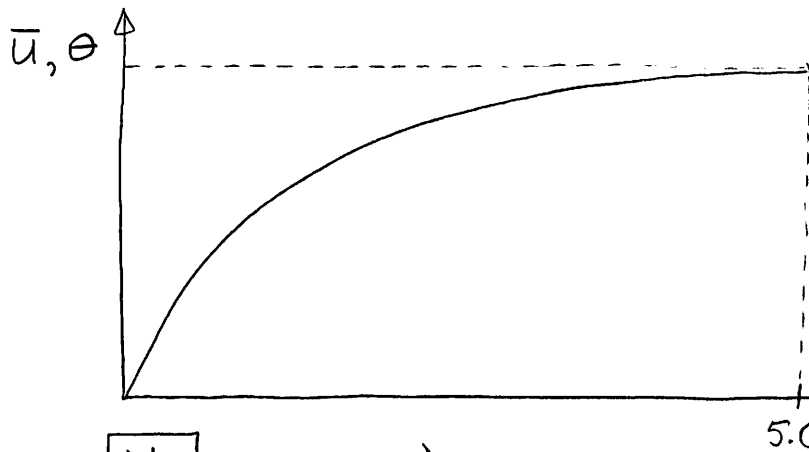
$$\frac{1}{Pr^{2/3}} \left(\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{2/3} \frac{\partial \theta}{\partial \eta^*} \right) = 0$$

$$\frac{\partial^2 \theta}{\partial \eta^{*2}} + \frac{1}{2} F(\eta^*) \frac{\partial \theta}{\partial \eta^*} = 0$$

Now our PDE becomes identical to before

$$\frac{\partial^2 \theta}{\partial \eta^{*2}} + \frac{1}{2} F(\eta^*) \frac{\partial \theta}{\partial \eta^*} = 0 \Rightarrow \theta(\eta^*) = \bar{u}(\eta)$$

B.C.'s: $\theta(\eta^*=0) = 0$, $\theta = \frac{T-T_0}{T_\infty-T_0}$
 $\theta(\eta^* \rightarrow \infty) = 1$



$$\delta \sqrt{\frac{V_\infty}{xU}} = \eta(y=\delta) = 5.0$$

$$Pr^{1/3} \delta_T \sqrt{\frac{V_\infty}{xU}} = \eta^*(y=\delta_T) = 5.0$$

$\eta, \eta^* = \eta Pr^{1/3}$
 $\hookrightarrow$ Thermal boundary layer
 $\hookrightarrow$ Hydrodynamic boundary layer.

Taking the ratio of our two boundary layer thicknesses:

$$\boxed{\frac{\delta}{\delta_T} = Pr^{1/3}} \Rightarrow \text{Makes sense, the only difference is } U \text{ and } \alpha$$

$Pr = \frac{\nu}{\alpha}$

Now we can solve for heat transfer

$$q''_{x=0} = -k \left. \frac{\partial T}{\partial y} \right|_{y=0} \Rightarrow \text{we know } \Theta = \frac{T-T_0}{T_\infty-T_0}, \quad \partial \Theta = \frac{\partial T}{T_\infty-T_0}$$

$$\begin{aligned} q''_{x=0} &= -k(T_\infty - T_0) \left(\frac{V_\infty}{xU} \right)^{1/2} Pr^{1/3} \left. \frac{\partial \Theta}{\partial \eta^*} \right|_{\eta^*=0} \\ &= \frac{k(T_0 - T_\infty)}{x} \underbrace{\left(\frac{V_\infty x}{U} \right)^{1/2}}_{Re_x^{1/2} \Rightarrow \text{Reynolds number}} Pr^{1/3} \underbrace{F''(0)}_{a_2 = 0.332} \end{aligned}$$

$\eta^* = \eta Pr^{1/3} = y \left(\frac{V_\infty}{xU} \right)^{1/2} Pr^{1/3}$
 $2\eta^* = 2y \left(\frac{V_\infty}{xU} \right)^{1/2} Pr^{1/3}$

$$q''_{x=0} = \frac{k \Delta T}{x} Re_x^{1/2} Pr^{1/3} a_2$$

$$Nu_x = \frac{hx}{k} = \underbrace{\frac{q''}{\Delta T}}_h \cdot \frac{x}{k} = a_2 Re^{1/2} Pr^{1/3} \Rightarrow \boxed{Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}}$$

Looking at shear stress, we showed before that:

$$C_{f,x} = \frac{\tau}{\frac{1}{2} \rho V_\infty^2} = \frac{2a_2}{\sqrt{Re_x}}, \text{ where } C_{f,x} = \text{skin friction coefficient}$$

$$\left(\frac{Nu_x}{Re \cdot Pr} \right) = \frac{1}{2} \left(\frac{2a_2}{\sqrt{Re_x}} \right) \frac{1}{Pr^{2/3}} = \frac{1}{2} C_{f,x} \frac{1}{Pr^{2/3}}$$

$$St = \frac{h}{\rho C_p V_\infty} \Rightarrow \text{Stanton Number}$$

$$St = \frac{\text{Heat transferred to a fluid}}{\text{Thermal capacity of the fluid}}$$

$\Rightarrow$ Characterizes heat transfer in forced convection flows.

We can write a general analogy that:

$$St \cdot Pr^{2/3} = \frac{C_{f,x}}{2}$$

$\Rightarrow$ Colburn Analogy
Colburn j-factor $\Rightarrow$

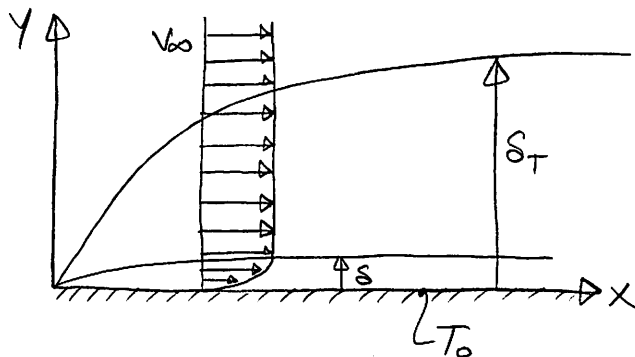
Aside: Another analogy
Ratio of $C_{f,x}$, and Nu_x

$$\frac{C_{f,x}}{2} = \frac{h_x}{\rho C_p V_\infty} Pr^{2/3} = j_H$$

This is a very powerful analogy relating heat, momentum, and mass transfer. Can relate heat transfer and temperature to shear & velocity, i.e. we can solve for shear & V_∞ by measuring q'' & T_o, T_∞ .

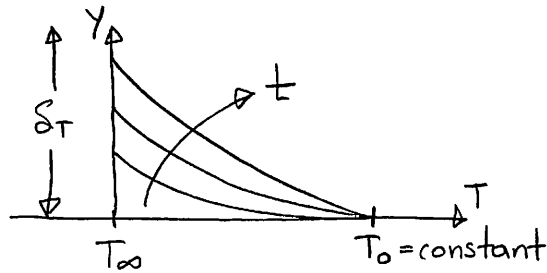
Some limits ($Pr \ll 1$)

If a fluid has a very low Prandtl number $\Rightarrow Pr = \frac{\nu}{\alpha} \ll 1$



$\Rightarrow$ Looks a lot like a transient conduction problem.

We can rethink this as:



$$q''_{x=0} = \frac{h \Delta T}{\sqrt{\pi \alpha t}} \Rightarrow \text{Solved before for semi-infinite conduction. Check Lecture 8, Page 66 of my notes.}$$

$$Nu_x = \left(\frac{q''}{\Delta T} \cdot \frac{x}{k} \right) = \frac{x}{\sqrt{\pi \alpha t}}$$

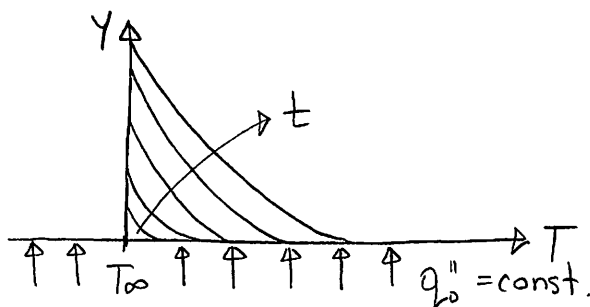
we know $t = \frac{x}{V_\infty}$

$$Nu_x = \frac{1}{\sqrt{\pi}} \cdot \left(\frac{V_\infty x}{\alpha} \right)^{1/2} \Rightarrow \boxed{Nu_x = \frac{1}{\sqrt{\pi}} \cdot Re_x^{1/2} Pr^{1/2}}$$

↳ For constant wall temperature, $T_0, Pr \ll 1$

Now if we have a constant heat flux: ($Pr \ll 1$)

$$q''_0 = \text{constant}$$



We know from our previous solution that: (Lecture 8, Page 67)

$$T - T_\infty = \frac{q''_0}{k} \left(\frac{4 \alpha t}{\pi} \right)^{1/2} e^{-\frac{x^2}{4 \alpha t}} - x \operatorname{erfc} \left(\frac{x}{\sqrt{4 \alpha t}} \right) \Rightarrow \text{Evaluate at } x=0$$

$$\underbrace{T_0 - T_\infty}_{\Delta T} = \frac{q''_0}{k} \left(\frac{4 \alpha t}{\pi} \right)^{1/2}$$

$$Nu_x = \left(\frac{q_0''}{\Delta T} \cdot \frac{x}{k} \right) = \frac{4}{\sqrt{\pi}} \cdot \frac{x}{\sqrt{\alpha t}} = \frac{4}{\sqrt{\pi}} \cdot \left(\frac{V_\infty x}{\alpha} \right)^{1/2}, \quad t = \frac{x}{V_\infty}$$

$$\boxed{Nu_x = \frac{4}{\sqrt{\pi}} \cdot Re^{1/2} Pr^{1/2}} \Rightarrow \text{Constant heat flux, } q_0'', \quad Pr \ll 1$$

Note that for these solutions, we've already defined $Pe = Re Pr$

$$Pe_x = \frac{V_\infty x}{\alpha} = \frac{\rho C_p V_\infty \Delta T}{k \Delta T} = \frac{\text{heat storage rate in the b.l.}}{\text{heat conductance through the b.l.}}$$

$$\boxed{Nu_x \propto Pe^{1/2}}$$

Average Heat Transfer Coefficient ($\bar{h}$)

$$\underbrace{\bar{h} = \frac{\bar{q}_L''}{\Delta T}}_{\text{Const. wall T.}} \quad \text{or} \quad \underbrace{\bar{h} = \frac{q''}{\overline{\Delta T}}}_{\text{Const. heat flux problems}}$$

Uniform Wall Temperature:

$$\bar{h} = \frac{\bar{q}_L''}{\Delta T} = \frac{1}{\Delta T} \left[\frac{1}{L} \cdot \int_0^L q'' dx \right] = \frac{1}{L} \int_0^L h(x) dx$$

Uniform Heat Flux

$$\bar{h} = \frac{q''}{\overline{\Delta T}} = \frac{q''}{\frac{1}{L} \int_0^L \Delta T(x) dx}$$

The Nusselt number based on $\bar{h}$ and L is $\overline{Nu}_L$

This is not the average of Nu_x .

For a flat plate case: ($x_0 = x(0) = 0$)

$$\bar{h} = \frac{1}{L} \int_0^L \underbrace{h(x)}_{\frac{k}{x} Nu_x} dx = \frac{0.332 k Pr^{1/3} \sqrt{V_\infty}}{L \sqrt{\nu}} \cdot \int_0^L \frac{\sqrt{x}}{x} dx = 0.664 Re_L^{1/2} Pr^{1/3} \left(\frac{k}{L} \right)$$

So now we see: $\bar{h} = 2h(x=L)$ in laminar flow

$$\boxed{\overline{Nu}_L = \frac{\bar{h}L}{k} = 0.664 Re_L^{1/2} Pr^{1/3}}$$

For $Pr \ll 1$:

$$\boxed{\overline{Nu}_L = 1.13 Pe_L^{1/2}}, \quad \boxed{Pe_L = Re_L Pr}$$

Some Observations and Notes

Previous results are valid under the following conditions:

- 1) Re_x or $Re_L < 5.0 \times 10^5$ (Laminar flow)
- 2) $Ma = \frac{V_\infty}{\text{sound speed}} < 0.3$ (Incompressible flow)
- 3) $Ec = \text{Eckert number} = \frac{V_\infty^2}{c_p(T_0 - T_\infty)} \ll 1$ (Viscous dissipation heating is negligible)
 $= \frac{\text{Kinetic Energy}}{\text{Enthalpy}}$

The higher the kinetic energy, the larger the effect of viscous dissipation.

- 4) We have always assumed that properties are constant. Need to evaluate properties at the average temperature of the boundary layer, or the film temperature:

$$T_f = \frac{T_0 + T_\infty}{2}$$

- 5) h or $\bar{h} \propto \frac{1}{\sqrt{x}}$ or $\frac{1}{\sqrt{L}}$, $Nu_x \propto \sqrt{x}$

Thus $h \rightarrow \infty$ and $Nu_x \rightarrow 0$ at $x \rightarrow 0$. Of course, $h \rightarrow \infty$ will not occur at $x \rightarrow 0$ since the b.l. model breaks down at $x=0$.

END OF LECTURE 10.